

Pedestrian-Centric Discriminative and Fine-grained Semantic Mining for Text-based Person Retrieval

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Abstract

Text-based Person Retrieval (TPR) aims to retrieve specific pedestrian images from a gallery based on the given textual descriptions, serving as a fine-grained instance of cross-modal retrieval on the Web. Current mainstream approaches primarily leverage pre-trained models and attention mechanisms to enhance multi-modal representations. Despite notable progress, they still struggle with two major challenges: 1) Intra-instance semantic asymmetry, which mainly derives from the partial semantic relevance conveyed by each image-text pair; and 2) Inter-instance semantic ambiguity, which arises from the high similarity of image-text pairs with different identities. These issues result in suboptimal semantic alignment and degraded retrieval accuracy. To this end, we propose a novel Pedestrian-Centric Discriminative and Fine-grained Semantic Mining (DFSM) framework for TPR. Specifically, our DFSM method comprises two essential components: 1) Text-aware Visual Refinement (TVR), which mitigates visual redundancy by selecting semantically relevant patches under textual guidance, and refines them via adaptive clustering and merging; 2) Token-level Semantic Alignment (TSA), which formulates the matching relationship between image regions and text words as a conditional transport (CT) problem, effectively mining fine-grained semantic differences and enhancing instance discrimination. Extensive experiments on four benchmarks validate the advantages of DFSM in terms of retrieval accuracy and visual interpretability.

CCS Concepts

• **Information systems** → **Information retrieval**; **Retrieval models and ranking**; **Similarity measures**.

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Keywords

Text-based Person Retrieval; Discriminative Information Mining; Fine-grained Alignment; Conditional Transport

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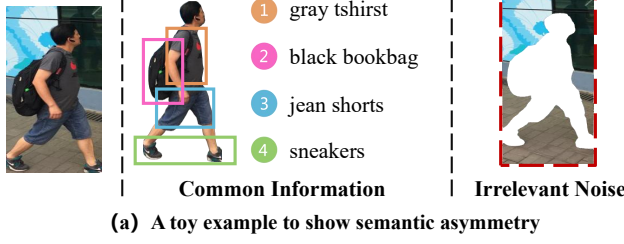
1 Introduction

Text-based person retrieval (TPR) [19] aims to retrieve the most semantically relevant pedestrian images based on the given textual descriptions. This task has broad applications in smart city infrastructure and security systems, but its significance goes beyond these domains. At its core, TPR represents a vertical and fine-grained instance of cross-modal retrieval, directly linking natural language queries with large-scale visual repositories. This formulation naturally connects TPR to broader challenges in Web search and retrieval, including query understanding, semantic matching across modalities, and effective ranking.

Towards this end, existing studies have proposed various strategies to enhance cross-modal alignment, including loss function design [28, 44], fine-grained modeling [38, 48], and pre-training strategies [12, 13, 33]. In addition, emerging approaches [2, 24, 27] propose interactive retrieval paradigms that progressively refine textual queries through multi-round dialogues.

Despite impressive performance, they still suffer from limitations in semantic representation and alignment bias, which constrain retrieval effectiveness. These issues are primarily reflected in two aspects: 1) Intra-instance semantic asymmetry, which results from the fact that the main content conveyed by each image-text pair is often partially related. As illustrated in Fig.1 (a), the text focuses on the man's salient appearance attributes (e.g., gray shirt, jean shorts, and backpack), while the corresponding image inevitably contains additional redundant information (e.g., irrelevant background or

Query: He is wearing a gray tshirt, jean shorts and sneakers. He has a black bookbag slung over his right shoulder.



This man is frowning and putting both hands in pockets. He wears a type of **red jacket**, **blue jeans** and **green sneakers**. A man in a **red down jacket**, **blue jeans** and dark sneakers is walking down the street and talking to someone next to him.



Figure 1: The motivation of DFSM. (a) Textual descriptions emphasize pedestrian features, while images contain irrelevant context, introducing noise. (b) Both visual content and textual descriptions of different IDs exhibit high similarity, introducing ambiguity in semantic alignment.

environmental context) not mentioned in the text. The redundant noise contaminates visual representations, which disrupts semantic alignment between modalities and ultimately weakens retrieval performance. 2) Inter-instance semantic ambiguity, stemming from the high similarity of image-text pairs with different identities. As illustrated in Fig.1 (b), the individuals of ID 2424 and 3492 exhibit nearly identical visual cues (e.g., red jacket and blue jeans), and the texts also convey similar semantic content. However, subtle differences still exist, such as the color of their sneakers. Without capturing the fine-grained distinctions, models relying solely on global matching may fail to distinguish visually similar identities, thereby significantly increasing the risk of mistaking negative instances as true matches. Collectively, these observations motivate the central research question of this study: “How can we effectively suppress pedestrian-irrelevant visual information and capture subtle fine-grained semantic differences to achieve robust TPR?”

To answer this question, we propose a novel Pedestrian-Centric Discriminative and Fine-grained Semantic Mining (DFSM) model, consisting of two key components: 1) **Text-aware Visual Refinement (TVR)**, which first effectively estimates the semantic relevance of visual patches and selects the most informative ones accordingly. Subsequently, we apply a density peaks clustering algorithm based on k-nearest neighbor [8] to adaptively group the selected patches, thereby enhancing spatial coherence and semantic completeness. The resulting clusters are further merged via the importance weighting. Finally, a dynamic fusion strategy is employed to refine the global representation, effectively reducing redundancy

and emphasizing discriminative features. 2) **Token-level Semantic Alignment (TSA)**, which models the fine-grained correspondences between visual regions and textual words through the conditional transport (CT) theory [45]. By minimizing the bidirectional CT costs between visual and textual distributions, TSA accentuates subtle semantic differences among similar instances, enhancing intra-class consistency while enlarging inter-class separation. Finally, we integrate the strengths of TVR and TSA to effectively suppress semantic redundancy and establish more reliable cross-modal alignment.

The main contributions are as follows:

- We propose a novel Pedestrian-Centric Discriminative and Fine-grained Semantic Mining (DFSM) framework, which addresses intra-instance semantic asymmetry and inter-instance semantic ambiguity problem in TPR.
- We introduce a Text-aware Visual Refinement module that adaptively enhances discriminative visual features by selecting and merging informative image patches.
- We design a Token-level Semantic Alignment module, which utilizes conditional transport theory to model fine-grained correspondences between images and texts.
- We demonstrate the superiority of our proposed method through extensive experiments and visualizations conducted on four widely used benchmark datasets.

2 Related Work

Text-based person retrieval plays a critical role in intelligent video surveillance and security systems. The primary challenge of TPR lies in accurately aligning textual descriptions with corresponding images. Early works [43, 46] tackle the problem by adopting global matching strategies, focusing on overall similarity between modalities. Recognizing that TPR is a fine-grained task, later works [10, 35, 36, 47] shift toward exploring local matching, broadly classified into explicit and implicit alignment strategies. Furthermore, inspired by the success of vision-language pretraining, recent studies [12, 20, 22, 40] have explored the use of CLIP [29] to strengthen representation learning. Nevertheless, the domain gap between pretraining data and TPR scenarios limits the performance of current methods. To address this, some works [13, 33] leverage multimodal large language models (MLLMs) to generate image captions and build extensive image-text pairs in the ReID domain for pretraining. Despite impressive performance, they ignore the intra-instance semantic asymmetry caused by irrelevant noise, which could hinder semantic alignment. This challenge serves as a key motivation for our research.

Token compression has been a hot topic in recent years, typically categorized into token pruning and token merging. The former removes less important tokens based on their importance. For example, DynamicViT [30] and AdaViT [25] incorporate a lightweight module to estimate token importance and discard less informative ones. However, such methods may lose useful information. To address this, EViT [21] and TPS [37] employ a new strategy, compressing the information from pruned tokens into a new token or reserved tokens. Conversely, token merging combines similar tokens into semantically richer units. Existing methods like LLaVA-PruMerge [31] and Dockylin [42] have explored the joint application of token pruning and merging. Inspired by this, we incorporate

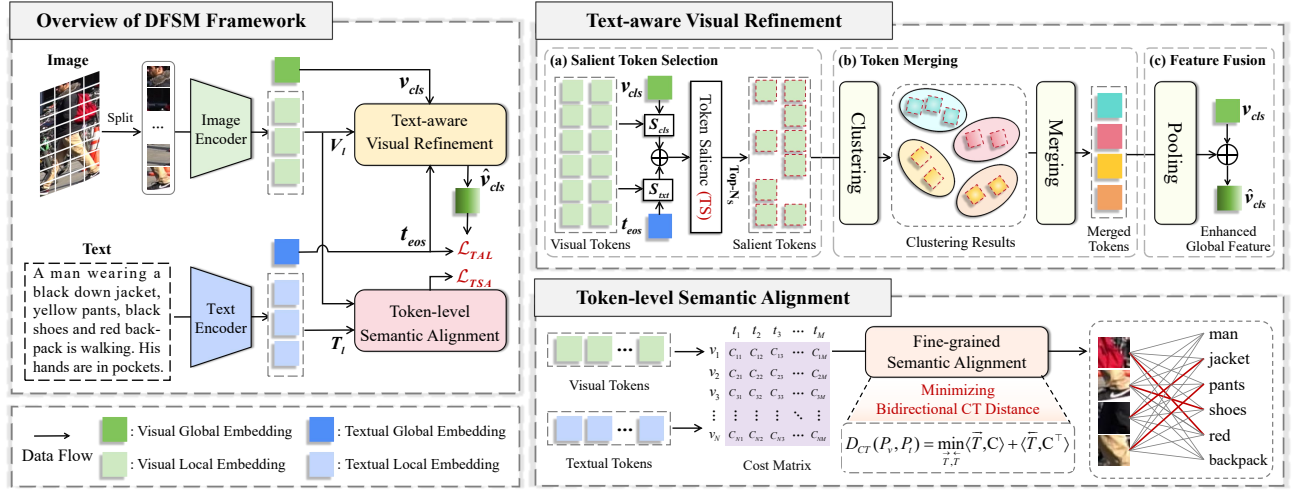


Figure 2: Overview of the proposed DFSM framework, which comprises two key components: the Text-aware Visual Refinement (TVR) module and the Token-level Semantic Alignment (TSA) module. The TVR progressively refines visual representation via a three-stage process, mitigating visual redundancy while enhancing feature completeness and consistency. The TSA module formulates region-word matching as a conditional transport problem to improve fine-grained semantic alignment.

token pruning and merging into the TPR, selecting informative tokens and then merging them to enable adaptive refinement, yielding more discriminative visual representations.

Conditional transport (CT) is an effective tool for quantifying the discrepancy between two probability distributions, and has recently gained increasing attention in various tasks such as multi-label image classification [17], few-shot learning [34], and compositional zero-shot learning [16]. Unlike Optimal Transport (OT) [26], which relies on Sinkhorn algorithms [6] to optimize the transport plan, CT [45] considers the transport plan using semantic similarity in a bidirectional view, offering advantages in computational efficiency and scalability. In this work, we incorporate CT theory to model fine-grained correspondences by minimizing the transport costs between image patches and words, thereby enhancing local feature alignment while suppressing redundant noise.

3 Methodology

3.1 Preliminary

The TPR task seeks to retrieve relevant images for a provided query. Let $D = \{(I_i, T_i)\}_{i=1}^N$ be the training set comprising N paired samples, where each pair consists of an image I_i and its associated text description T_i . For an image-text pair (I, T) , we utilize the visual encoder to generate patch embeddings $V = \{v_{cls}, v_1, \dots, v_N\}$, where v_{cls} denotes the global visual feature, and $V_l = \{v_1, \dots, v_N\}$ represents the local features. Similarly, the text encoder processes T into token embeddings $T = \{t_{sos}, t_1, \dots, t_{eos}\}$, where t_{eos} is used as the global text feature, and t_{sos} serves as a non-semantic start anchor that initializes and marks the beginning of the text sequence. The local text features are represented as $T_l = \{t_1, \dots, t_M\}$. Finally, the cosine similarity is employed to quantify the correspondence between the visual and textual features. Accurate semantic alignment across modalities is critical for effective retrieval. However,

as shown in Fig.1, achieving optimal alignment is challenging due to biases in semantic representation and feature alignment. Therefore, our research focuses on two key challenges: (1) how to precisely extract discriminative visual features, and (2) how to improve fine-grained matching to distinguish highly similar instances. The overview of DFSM is presented in Fig.2, which consists of two main components described in the following sections.

3.2 Text-aware Visual Refinement

Intra-instance semantic asymmetry caused by partial semantic relevance within image-text pairs hinders effective cross-modal alignment. To mitigate this, we propose a Text-aware Visual Refinement (TVR) module that effectively eliminates redundant patches via the textual guide, and then aggregates the informative patches into continuous visual attribute representations to refine the semantic and structural consistency. There are three sequential processes: Salient Token Selection, Token Merging, and Feature Fusion.

Salient Token Selection. Inspired by prior works [30, 37], we regard token selection as a classification task that estimates a saliency score for each token and selects the most informative ones for subsequent refinement. To this end, we design two complementary relevance scores to guide the selection.

① *Class attention score* is derived from the self-attention map in the final Transformer layer, which quantifies the correlation between the [CLS] token and other visual tokens. Given the attention map $A \in \mathbb{R}^{(1+N) \times (1+N)}$ of image I , it can be defined as:

$$S_{cls}^i = \{a_i\}_{i=1}^N = A[0, 1 : N + 1] \in \mathbb{R}^N, \quad (1)$$

② *Text-related score* captures text-guided semantic relevance by measuring the association between each visual token v_i and the global textual feature t_{eos} . Mathematically, it can be expressed as:

$$S_{txt}^i = \frac{\exp(v_i^\top t_{eos})}{\sum_{j=1}^N \exp(v_j^\top t_{eos})}, \quad i = 1, \dots, N, \quad (2)$$

where the softmax operation normalizes the similarity scores into a probabilistic distribution across tokens. Finally, we combine the two scores to obtain the overall token saliency (TS) score:

$$TS(i) = (S_{cls}^i + S_{txt}^i)/2. \quad (3)$$

This formulation yields a more comprehensive and reliable estimate of token importance. Next, we select the top- N_s tokens based on their saliency scores as the salient token sequence $V_s = \{v_s^i\}_{i=1}^{N_s}$. The remaining $N - N_s$ tokens are categorized as the redundant token sequence $V_r = \{v_r^i\}_{i=1}^{N-N_s}$, where $N_s = R_s N$ and R_s denotes the token selection rate.

Token Merging. Although the selected tokens exhibit high discriminative relevance to the pedestrian identity, they remain scattered and insufficient to represent complete attribute concepts. To this end, we propose a token merging strategy to improve the semantic completeness and structural consistency of visual representations. Specifically, we first cluster the salient tokens into a fixed number of groups, with each cluster expected to capture a distinct attribute-level concept (e.g., clothing or items). Subsequently, the tokens within each cluster are aggregated into a new token via the importance weighting. This process effectively transforms scattered local features into unified attribute tokens, facilitating subsequent fine-grained alignment.

For clustering, we adopt a density peak clustering algorithm based on k-nearest neighbor (DPC-KNN) [8]. Specifically, given the selected tokens V_s , we first estimate the local density ρ_i of each token v_s^i by averaging its distance to the k-nearest neighbors, formulated as:

$$\rho_i = \exp\left(-\frac{1}{K} \sum_{v_s^j \in \text{KNN}(v_s^i)} \|v_s^i - v_s^j\|^2\right), \quad (4)$$

where $\text{KNN}(v_s^i)$ denotes the k-nearest neighbors of v_s^i . Then, we compute the distance indicator δ_i for each token, which is defined as the minimum distance to any other token with higher local density:

$$\delta_i = \begin{cases} \min_{j: \rho_j > \rho_i} \|v_s^i - v_s^j\|^2, & \text{if } \exists j \text{ s.t. } \rho_j > \rho_i. \\ \max_j \|v_s^i - v_s^j\|^2, & \text{otherwise.} \end{cases} \quad (5)$$

Next, we consider the top- N_m tokens as cluster centers according to the values of $\rho_i \times \delta_i$, and then assign the remaining tokens to the nearest cluster based on Euclidean distance. The N_m is determined by the token merging rate R_m , i.e., $N_m = R_m N_s$.

Since each cluster may contain a varying number of tokens with different importance, we adopt a saliency-guided weighted sum to emphasize more informative tokens, where the weight s_j is inherited from the token saliency (TS) score introduced in the Salient Token Selection stage and normalized again within each cluster. Formally, the merged token v_m^i for the i -th cluster C_i is computed as:

$$v_m^i = \sum_{j \in C_i} \frac{s_j}{\sum_{k \in C_i} s_k} \cdot v_s^j, \quad (6)$$

where v_s^j denotes the j -th token in the cluster, and s_j is its saliency score. All merged tokens are then concatenated to form the refined visual representation $V_m = \{v_m^i\}_{i=1}^{N_m}$.

Feature Fusion. To further enhance global representations after token refinement, we propose a feature fusion strategy. Specifically,

we first apply the Generalized Pooling Operator (GPO) [5] to produce an enhanced global feature $\bar{v}_{cls} = \text{GPO}(V_m)$. Afterwards, we fuse $\bar{v}_{cls}$ with v_{cls} via a balance coefficient. The process is defined as follows:

$$\hat{v}_{cls} = \lambda_{cls} \cdot v_{cls} + (1 - \lambda_{cls}) \cdot \bar{v}_{cls}, \quad (7)$$

where λ_{cls} is a learnable coefficient. In this way, we obtain the final visual representation $\hat{v}_{cls}$ that contains both contextual semantics and enhanced discriminative properties.

3.3 Token-level Semantic Alignment

Although global features enable a holistic view of cross-modal alignment, they often neglect subtle local cues that are crucial for distinguishing subtle differences across instances. This limitation is particularly pronounced in scenarios with high inter-instance similarity, where relying solely on global information matching may lead to retrieval ambiguity. Recent studies [12, 18] have attempted to address this challenge through implicit reasoning to achieve local alignment. While notable improvements, they remain sensitive to visual noise and suffer from instability caused by random masking strategies [15]. To overcome the issue, we propose a Token-level Semantic Alignment (TSA) module, which explicitly formulates the fine-grained alignment between image regions and text words as the Conditional Transport (CT) problem. By optimizing the bi-directional transport plan, TSA captures fine-grained semantic correspondences between visual regions and textual tokens, improving retrieval performance in ambiguous scenarios. Given the visual tokens $V_l = \{v_i\}_{i=1}^N \in \mathbb{R}^{N \times d}$ and the words features $T_l = \{t_j\}_{j=1}^M \in \mathbb{R}^{M \times d}$, we define two discrete distributions P_v and P_t as follows:

$$P_v = \sum_{i=1}^N a_i \delta_{v_i}, P_t = \sum_{j=1}^M b_j \delta_{t_j}, \quad (8)$$

where δ_{v_i} and δ_{t_j} represent Dirac delta functions centered at the feature points v_i and t_j , respectively. $\mathbf{a} = \{a_i\}_{i=1}^N \in \Delta_N$ and $\mathbf{b} = \{b_j\}_{j=1}^M \in \Delta_M$, with Δ representing the probability simplex of N - and M -dim. In practice, we employ the uniform distribution to initialize the weights, i.e., $a_i = 1/N$ and $b_j = 1/M$. Accordingly, the CT distance between μ and ν is defined from two directions:

$$D_{CT}(P_v, P_t) = \min_{T^{v \rightarrow t}, T^{t \rightarrow v}} \langle T^{v \rightarrow t}, C \rangle + \langle T^{t \rightarrow v}, C^\top \rangle \quad (9)$$

$$\text{s.t. } T^{v \rightarrow t} \mathbf{1}_M = \mathbf{a}, \quad T^{t \rightarrow v} \mathbf{1}_N = \mathbf{b},$$

where $\langle \cdot, \cdot \rangle$ denotes the Frobenius inner product. $C \in \mathbb{R}^{N \times M}$ is the transport cost matrix, with each element computed as $C_{ij} = 1 - \cos(v_i, t_j)$. The forward transport plan $T^{v \rightarrow t}$ captures the soft matching from the visual tokens to the words, and is defined as:

$$T_{ij}^{v \rightarrow t} = a_i \cdot \frac{b_j \exp(\cos(v_i, t_j))}{\sum_{j'=1}^M b_{j'} \exp(\cos(v_i, t_{j'}))}, \quad (10)$$

This formulation encourages each visual token to transport more mass to the semantically relevant word tokens. Similarly, the backward transport plan $T^{t \rightarrow v}$, which aligns word tokens to their most semantically relevant visual tokens, is defined as:

$$T_{ji}^{t \rightarrow v} = b_j \cdot \frac{a_i \exp(\cos(t_j, v_i))}{\sum_{i'=1}^N a_{i'} \exp(\cos(t_j, v_{i'}))}. \quad (11)$$

Table 1: Performance comparisons with SOTA methods on the CUHK-PEDES, ICFG-PEDES, and RSTPReid datasets. The results of NAM and HAM are reproduced by us, where each method is fine-tuned on the IRRa [12] using their respective publicly released pre-trained weights. “-” means that the result is not reported by the paper.

Method	Ref	CUHK-PEDES				ICFG-PEDES				RSTPReid			
		R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP
LCR ² S [39]	MM23	67.36	84.19	89.62	59.20	57.93	76.08	82.40	38.21	54.95	76.65	84.70	40.92
CFine [40]	TIP23	69.57	85.93	91.15	-	60.83	76.55	82.42	-	50.55	72.50	81.60	-
IRRA [12]	CVPR23	73.38	89.93	93.71	66.13	63.46	80.25	85.82	38.06	60.20	81.30	88.20	47.17
TBPS [3]	AAAI24	73.54	88.19	92.35	65.38	65.05	80.34	85.47	39.83	62.10	81.90	87.75	48.00
UMSA [44]	AAAI24	74.25	89.83	91.58	66.15	65.62	80.54	85.83	38.78	63.40	83.30	90.30	49.28
IRLT [23]	AAAI24	74.46	90.19	94.01	-	64.72	81.35	86.31	-	61.49	82.26	89.23	-
DCEL [20]	MM23	75.02	90.89	94.52	-	64.88	81.34	86.72	-	61.35	83.95	90.45	-
RDE [28]	CVPR24	75.49	90.19	94.12	67.56	67.68	82.47	87.36	40.06	65.35	83.95	89.90	50.88
CFAM [48]	CVPR24	75.60	90.53	94.36	67.27	65.38	81.17	86.35	39.42	62.45	83.55	91.10	49.50
RaSa [1]	IJCAI23	76.51	90.29	94.25	69.38	65.28	80.40	85.12	41.29	66.90	86.50	91.35	52.31
APTM [41]	MM23	76.53	90.34	94.65	69.31	68.51	82.09	87.56	41.22	67.50	85.70	91.45	52.56
NAM [33]	CVPR24	77.00	91.20	94.59	69.21	67.26	82.45	87.24	41.69	68.65	86.40	91.80	52.77
AUL [18]	AAAI24	77.23	90.43	94.41	-	69.16	83.32	88.37	-	71.65	87.55	92.05	-
HAM [13]	CVPR25	77.66	91.36	94.70	69.16	67.89	83.19	87.86	41.78	71.65	87.50	92.75	55.09
ICL [27]	CVPR25	78.18	91.63	94.83	69.58	69.22	83.49	88.06	42.34	70.00	86.60	91.70	54.16
DFSM (Ours)	-	79.03	92.13	95.35	70.18	69.88	84.30	88.81	41.88	73.60	88.00	93.00	54.92

Overall, the bidirectional transport plan facilitates mutual guidance between visual and textual modalities, thereby promoting precise token-level alignment and enhancing the ability to distinguish subtle semantic differences.

3.4 Overall Objective

Our training objective comprises two components: **1)** We employ the Triplet Alignment Loss [28] ($\mathcal{L}_{TAL}$) to provide deterministic supervision at the global level. **2)** We further optimize fine-grained matching using the proposed $\mathcal{L}_{TSA} = \sum_{i=1}^B D_{CT}$. The B denotes the batchsize. The total loss is given by:

$$\mathcal{L}_{Total} = \mathcal{L}_{TAL} + \alpha \mathcal{L}_{TSA}, \quad (12)$$

where α is a trade-off hyperparameter.

4 Experiments

4.1 Experimental Settings

Datasets. We evaluate our method on four benchmarks, including three coarse-grained datasets CUHK-PEDES [19], ICFG-PEDES [7], RSTPReid [47], and one fine-grained dataset UFine6926[48]. For all datasets, we adopt the official data partitioning protocols. The introduction of each dataset is provided in the Appendix.

Evaluation Metrics. For all experiments, we utilize the R@ k metrics ($k=1,5,10$) and mean Average Precision (mAP) for evaluation.

Implementation Details. All experiments are conducted on a single RTX3090 24GB GPU. We adopt CLIP-ViT-B/16 as the backbone and use the weights released by [13] for parameter initialization. The images are resized to 384×128 . The maximum text length is 77 (168 for UFine6926) following [48]. We train DFSM for 30 epochs using the Adam optimizer [14], with an initial learning rate of 1×10^{-5} progressively reduced by a cosine decay scheduler. The

batch size is 128, except for UFine6926, where it is set to 112. The hyperparameters R_s is set to 0.5 for CUHK-PEDES and RSTPReid, and 0.6 for ICFG-PEDES and UFine6926. The hyperparameters R_m is set to 0.3 for CUHK-PEDES, RSTPReid, and UFine6926, and 0.2 for ICFG-PEDES. The hyperparameters α in Eq.(12) is set to 1.0.

4.2 Comparison with Competitive Methods

Results on coarse-grained datasets. Tab.1 reports the results on the three coarse-grained benchmarks, CUHK-PEDES, ICFG-PEDES, and RSTPReid. We have the following observations:

- **On CUHK-PEDES**, DFSM consistently achieves state-of-the-art performance on all evaluation metrics. Specifically, DFSM exhibits the improvement of +1.37% in terms of R@1 accuracy over HAM [13], which fine-tunes IRRA [12] and initializes parameters by the same pre-training weights as us. These results highlight the capability of DFSM to effectively extract meaningful pedestrian features and capture fine-grained relationships, thereby achieving outstanding retrieval performance.
- **On ICFG-PEDES**, DFSM attains 69.88%, 84.30%, and 88.81% in terms of R@1, R@5, and R@10 accuracy, respectively, outperforming all existing methods. This supports the idea that improving the discriminative power of representations is advantageous for cross-modal alignment. Although our model shows a slight disadvantage in mAP compared to ICL [27], it achieves higher recall at multiple levels, reflecting its superior overall retrieval performance.
- **On RSTPReid**, DFSM shows clear advantages in R@1 over the others. Despite the relatively small scale of this dataset, our method surpasses AUL [18] by +1.95% in R@1 accuracy, highlighting its robustness against overfitting and its effectiveness for text-based person retrieval in low-resource scenarios.

Table 2: Performance comparisons on the generalization capabilities between coarse-grained datasets CUHK-PEDES, ICFG-PEDES and RSTPReid.

Methods	Training Set	CUHK-PEDES				ICFG-PEDES				RSTPReid			
		R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP
APTM [41]	CUHK-PEDES	76.53	90.34	94.65	69.31	46.20	65.13	72.59	22.66	54.25	78.05	85.25	39.17
	ICFG-PEDES	48.67	68.75	77.06	36.83	68.51	82.09	87.56	41.22	54.65	76.50	83.90	38.20
	RSTPReid	57.09	76.52	83.72	34.47	33.24	51.18	58.63	20.82	67.50	85.70	91.45	52.56
NAM [33]	CUHK-PEDES	77.00	91.20	94.59	69.21	50.73	68.69	75.64	28.06	59.40	80.50	88.05	44.73
	ICFG-PEDES	61.11	80.69	87.12	55.64	67.26	82.45	87.24	41.69	55.30	77.35	85.60	40.82
	RSTPReid	61.37	80.33	86.39	55.79	49.04	65.65	72.46	30.91	68.65	86.40	91.80	52.77
HAM [13]	CUHK-PEDES	77.66	91.36	94.70	69.16	59.47	76.54	82.67	35.10	58.00	79.20	87.35	43.62
	ICFG-PEDES	69.89	86.71	91.61	62.71	67.89	83.19	87.86	41.78	57.35	78.80	87.00	43.36
	RSTPReid	68.94	86.48	91.14	62.36	58.92	75.10	82.20	34.71	71.65	87.50	92.75	55.09
DFSM (Ours)	CUHK-PEDES	79.03	92.13	95.35	70.18	61.19	77.86	83.68	35.36	62.05	83.30	89.65	46.63
	ICFG-PEDES	72.53	88.29	92.65	64.97	69.88	84.30	88.81	41.88	62.80	83.05	88.65	46.30
	RSTPReid	71.49	86.96	91.82	64.16	60.03	75.49	84.39	36.44	73.60	88.00	93.00	54.92

Table 3: Performance comparisons with other methods on UFine6926 dataset.

Methods	Ref	R@1	R@5	R@10	mAP
NAFS[11]	arXiv21	64.11	80.32	85.05	63.47
LGUR[32]	MM22	70.69	84.57	89.91	68.93
SSAN[7]	arXiv21	75.09	88.63	92.84	73.14
IRRA[12]	CVPR23	85.02	94.31	96.75	83.91
CFAM(B/16)[48]	CVPR24	85.55	94.51	97.02	84.23
RDE [28]	CVPR24	87.60	95.65	97.46	86.10
CFAM(L/14)[48]	CVPR24	88.51	95.58	97.49	87.09
NAM [33]	CVPR24	89.65	96.28	97.82	88.35
HAM [13]	CVPR25	89.92	96.24	97.98	88.62
ICL [27]	CVPR25	91.02	96.98	98.17	89.76
DFSM (Ours)	-	91.67	97.31	98.40	89.84

Results on fine-grained dataset. We further evaluate DFSM on the fine-grained UFine6926 benchmark, with the results summarized in Tab.3. By observing Tab.3, we can see that:

- DFSM achieves remarkable improvements on UFine6926, reaching 91.67% in R@1 accuracy and 89.84% in mAP, and clearly surpasses other SOTA methods. Notably, our model achieves 3.16% improvements in R@1 over CFAM (L/14), which is explicitly designed for fine-grained alignment and matching, demonstrating the superior capability of our approach in capturing subtle cross-modal correspondences.
- Moreover, the performance on the fine-grained UFine6926 dataset is notably higher compared to the coarse-grained benchmarks. We attribute this to the longer and more detailed textual descriptions in UFine6926 (average length > 80), which provide richer semantic cues. These detailed descriptions enable our method to better leverage its fine-grained modeling capacity, resulting in improved performance across various metrics.

Table 4: Performance comparisons on the generalization capabilities between fine-grained and coarse-grained datasets.

Source → Target	Method	R@1	R@5	R@10	mAP
Ufine. → CUHK.	NAM [33]	54.03	74.43	82.42	48.77
	HAM [13]	55.70	76.25	83.25	50.31
	Our DFSM	60.38	79.34	85.85	54.17
Ufine. → ICFG.	NAM [33]	46.19	64.53	72.52	24.85
	HAM [13]	48.94	68.05	75.34	26.82
	Our DFSM	52.18	70.35	77.15	28.20
Ufine. → RSTP.	NAM [33]	51.75	74.80	84.60	38.62
	HAM [13]	53.40	76.05	84.45	39.25
	Our DFSM	57.50	79.55	86.75	42.16

4.3 Generalization Study

To better assess the generalization ability of DFSM, we perform cross-domain experiments on four benchmark datasets, with the results summarized in Tab.2 and Tab.4. As shown, DFSM consistently outperforms existing methods by a substantial margin. For instance, in the ICFG.→CUHK. setting (see Tab.2), DFSM outperforms NAM [33] and HAM [13] by 11.42% and 2.64% in R@1 accuracy, respectively, with similar gains observed in other settings. We attribute this notable improvement to two main factors: (1) Most methods overlook the hindering impact of redundant noise on cross-modal alignment. In contrast, DFSM explicitly reduces such interference by learning more discriminative representations for improved alignment. (2) DFSM leverages conditional transport theory to model fine-grained semantic relationships between modalities. This mechanism enriches the representation with more detailed semantic information, allowing the model to generalize beyond identity-specific features and better handle unseen domains. Overall, these results on the domain generalization task demonstrate the strong generalization capacity of DFSM across diverse datasets.

Table 5: Ablation study on ICFG-PEDES and RSTPReid. TM denotes Token Merging. The baseline refers to fine-tuning CLIP with the TAL loss.

Methods	ICFG-PEDES			RSTPReid		
	R@1	R@5	R@10	R@1	R@5	R@10
Different Components						
Baseline	66.73	82.16	87.17	70.10	86.95	91.30
w/o TVR	68.81	83.39	88.24	72.40	87.25	92.35
w/o TSA	69.00	83.52	88.37	72.90	88.10	92.80
DFSM	69.88	84.30	88.81	73.60	88.00	93.00
Different Settings in TVR						
TVR	69.00	83.52	88.37	72.90	88.10	92.80
w/o S_{txt}	68.37	82.91	87.70	72.54	87.45	92.70
w/o TM	68.12	82.76	87.68	72.05	87.45	92.50

Table 6: Comparison of Conditional Transport (CT) and Optimal Transport (OT) for the TSA Module on RSTPReid dataset.

Method	Flops(G)	Time(min) Per Batch	R@1	R@5	R@10
TSA (OT)	20.306	3.052	72.00	86.78	92.02
TSA (CT)	20.301	2.858	72.40	87.25	92.35

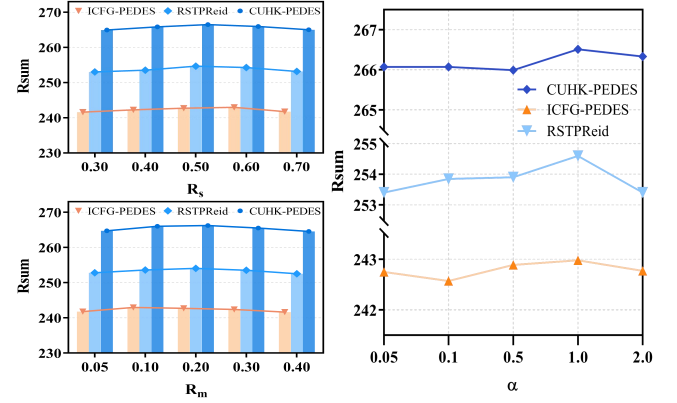
4.4 Ablation Study and Further Discussion

Effect of Text-aware Visual Refinement (TVR). From Tab.5, removing TVR (*i.e.*, w/o TVR) causes a clear performance drop, highlighting the importance of enhancing visual discriminability for cross-modal alignment. We further examine different settings in TVR. When S_{txt} is disabled (*i.e.*, relying on S_{cls} solely to assess token importance), performance declines, validating the importance of incorporating text-relevant relevance estimation. Similarly, removing the TM module (*i.e.*, applying Feature Fusion operation directly to selected tokens) also results in performance degradation, revealing the necessity of refining discrete tokens for better representation.

Effect of Token-level Semantic Alignment (TSA). As shown in Tab.5, removing the TSA module (*i.e.*, w/o TSA) leads to a decrease of 0.88% and 0.7% in R@1 accuracy on the ICFG-PEDES and RSTPReid dataset, respectively. The findings demonstrate that TSA facilitates fine-grained matching, which in turn enables the model to capture subtle local distinctions among pedestrians.

Analysis of Different Modeling Approaches (CT vs OT). To demonstrate the advantages of our introduced CT theory for fine-grained modeling, we provide an in-depth comparison with Optimal Transport (OT) theory in Tab.6. In contrast to OT’s reliance on iterative optimization (*e.g.*, the Sinkhorn algorithm) to derive the transport plan, CT considers the problem bidirectionally based on semantic similarity between samples from two distributions. We extensively compare them under our proposed TSA module, and observe a slight but consistent performance gain on all Rank-k metrics. Notably, our major advantage is the computational efficiency.

Specifically, TSA(CT) offers better runtime efficiency, reducing average training time per batch from 3.052 to 2.858 minutes.

**Figure 3: Variation of performance with different R_s , R_m , and α . We employ Rsum to measure overall quality, which is defined as the sum of R@1, R@5, and R@10. All variants of the R_s and R_m are evaluated without the TSA module.**

4.5 Hyper-parameter Analysis

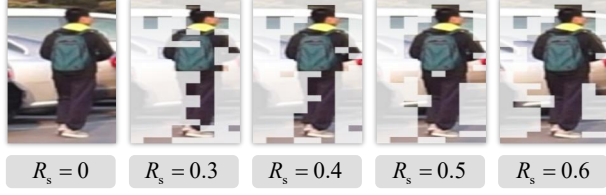
The hyperparameters R_s and R_m control the number of selected and merged tokens, respectively, and thus play a critical role in shaping pedestrian-centric representations. In addition, the balancing hyperparameter α in Eq.(12) adjusts the relative contribution of the global-level alignment loss $\mathcal{L}_{TAL}$ and the token-level alignment loss $\mathcal{L}_{TSA}$. To evaluate the sensitivity of DFSM to these hyperparameters, we conduct ablation studies across three benchmarks. As shown in Fig.3, both excessively low and high values of R_s or R_m lead to performance degradation, either by suppressing expressive information or amplifying irrelevant noise. In contrast, DFSM remains stable over a wide range of α values, with only minor performance fluctuations observed. These findings demonstrate the robustness and general applicability of our method under varying hyperparameter configurations.

4.6 Qualitative Analysis

To intuitively evaluate the effectiveness of DFSM, we provide qualitative analyses with examples presented in Fig.4-6.

Visual Token Selection Visualization. In Fig.4, we present the visualization of token selection with various selection ratios R_s on the RSTPReid dataset. The figures include textual descriptions, original images, and their corresponding processed outputs, where masked regions highlight tokens identified as redundant and subsequently discarded. These visualization results are consistent with the hyper-parameter analysis in Fig.3, confirming that overly small selection ratios discard key information whereas large selection ratios preserve redundant content. With appropriate token selection ratios, DFSM effectively retains relevant regions while filtering out irrelevant ones. For instance, as shown in Fig.4 (left), DFSM removes the background “car” at $R_s = 0.4$ and 0.5 , which is irrelevant to the query. This further highlights DFSM’s interpretability in handling redundant visual representations.

The young boy wears a black sport top, dark blue trousers and grey shoes. He takes a dark green bag. He is walking.



A woman with a pair of white earmuffs is walking on road. She wears a purple down jacket, a pair of blue jeans and pink trainers.

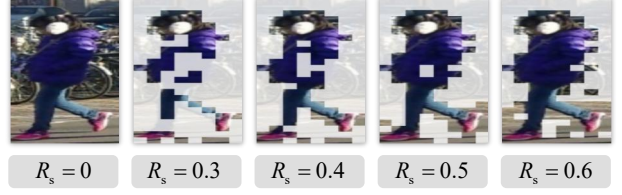
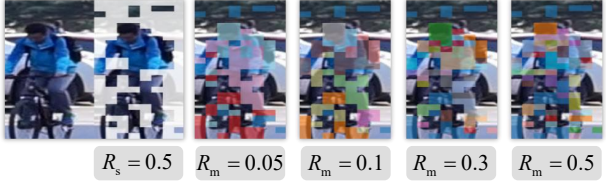


Figure 4: The visualization of selected tokens with associated texts under different selection ratios.

The boy is wearing a blue **coat**, grey **trousers** and sports **shoes**. He was carrying a gray **backpack**. He is riding a **bicycle**.



A woman with long black **hair** is wearing a red **coat**, blue **jeans** and black **shoes**. She is also carrying a blue **backpack**.

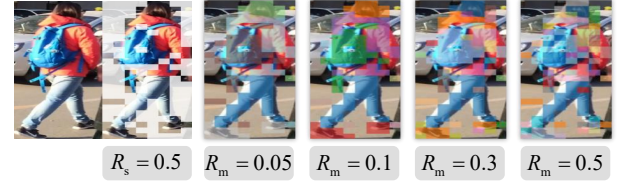
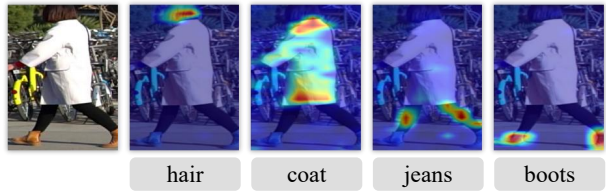


Figure 5: The visualization of merged tokens with the different merging ratios.

A woman with sholder length **hair** is walking. She is wearing a **white coat**, a pair of **yellow short boots** and **black jeans**.



This woman has long **black hair** and she is wearing a **pink sweater**, **Gray pants** and a **white backpack**.

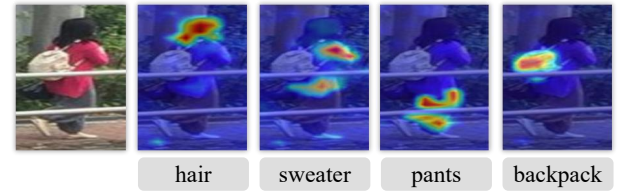


Figure 6: The visualization of region-word alignment. Highlighted visual regions indicate relevance to the words.

Visual Token Merging Visualization. Fig.5 illustrates the merged tokens under different merging ratios R_m . For clarity, clustered tokens are visualized as color-coded masks, where each color denotes a group identified by the DPC-KNN algorithm. Obviously, DFSM can effectively cluster attribute tokens of different visual scales (e.g., backpack and shoes) under appropriate merging ratios, enhancing the completeness of semantic representation. This reflects its ability to maintain semantic coherence while adapting diverse granularity levels. In contrast, higher R_m often results in overly coarse clusters, obscuring the semantic information.

Region-Word Alignment Visualization. We further evaluate the effectiveness of DFSM in modeling fine-grained associations by visualizing heatmaps via the gradient-weighted attention [4], where colors closer to red reflect more semantic relevance. As shown in Fig.6, textual attributes such as “coat” and “jeans” are accurately mapped to their corresponding visual regions. It is worth noting that DFSM successfully aligns visual and textual information even for small-scale attributes like “shoes”. These findings suggest that its fine-grained alignment capability not only effectively resolves inter-instance semantic ambiguity but also reduces the risk of overfitting.

5 Conclusion

In this paper, we propose a novel Pedestrian-Centric Discriminative and Fine-grained Semantic Mining (DFSM) framework to address the challenges of semantic asymmetry and ambiguity. DFSM integrates the Text-aware Visual Refinement (TVR), which selects semantically relevant tokens and enhances visual representations through adaptive clustering and merging. Additionally, the Token-level Semantic Alignment (TSA) module is introduced to model fine-grained relational matching via conditional transport theory, thereby improving the model’s ability to distinguish visually similar instances. Extensive experiments on four benchmarks demonstrate the superiority of DFSM in both retrieval performance and visual interpretability. Future research could focus on developing more adaptive or learnable selection mechanisms to improve flexibility and generalization capability across diverse scenarios.

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Appendix

In this Appendix, we further provide the following descriptions and visualizations to elaborate on the results and conclusions presented in the main paper.

- **Sec. A:** Detailed Explanation of the $\mathcal{L}_{TAL}$ Loss.
- **Sec. B:** Detailed Introduction of Datasets.
- **Sec. C:** More Qualitative Analysis.

A Detailed Explanation of the $\mathcal{L}_{TAL}$ Loss

To ensure more stable and comprehensive optimization for global alignment, we employ a novel Triplet Alignment Loss (TAL) [28], which alleviates the instability problem caused by over-optimization of the hardest negative samples during training with the traditional Triplet Ranking Loss (TRL) [9] in the Text-based Person Retrieval (TPR) task. For an input pair (I_i, T_i) , the $\mathcal{L}_{TAL}$ loss is defined as:

$$\begin{aligned} \mathcal{L}_{TAL} = & \sum_{i=1}^B [m - S_{i2t}^+(I_i) + \gamma \log(\sum_{j=1}^B (1 - m_{ij}) \exp(S(I_i, T_j)/\gamma))] + \\ & + \sum_{i=1}^B [m - S_{i2t}^+(T_i) + \gamma \log(\sum_{j=1}^B (1 - m_{ji}) \exp(S(I_j, T_i)/\gamma))] +, \end{aligned} \quad (13)$$

where B is the batch size, m is the margin parameter, and γ is the temperature hyperparameter. m_{ij} represents the matching label: $m_{ij} = 1$ indicates that (I_i, T_j) is a matched pair from the same identity, while $m_{ij} = 0$ indicates an unmatched pair. $[x]_+ = \max(x, 0)$, and $\exp(x) = e^x$. Given that one batch of data may contain multiple image-text pairs from the same identity, a weighted averaging strategy is adopted to calculate the similarity of positive samples:

$$S_{i2t}^+(I_i) = \sum_{j=1}^B a_{ij} S(I_i, T_j), \quad (14)$$

$$a_{ij} = \frac{m_{ij} \exp(S(I_i, T_j)/\gamma)}{\sum_{j=1}^B m_{ij} \exp(S(I_i, T_k)/\gamma)}. \quad (15)$$

The same formulation applies symmetrically for $S_{i2t}^+(T_i)$ in the reverse direction. By softly aggregating multiple positives and relaxing the negative pressure, $\mathcal{L}_{TAL}$ improves both convergence and retrieval quality in practice.

B Detailed Introduction of Datasets

We evaluate DFSM on four widely used TIREID benchmarks: CUHK-PEDES [19], ICFG-PEDES [7], RSTPReid [47] and UFine6926[48], to verify its effectiveness and superiority. The detailed introduction of these datasets is provided as follows:

- **CUHK-PEDES** [19] is the first large-scale pedestrian retrieval dataset with textual annotations. It contains 40,206 images and 80,412 textual descriptions corresponding to 13,003 identities.

- **ICFG-PEDES** [7] is a recent large-scale TPR dataset containing 54,522 images of 4,102 unique persons. Each image is paired with one textual description.
- **RSTPReid** [47] comprises 20,505 images of 4,101 individuals collected from 15 cameras, and each image is annotated with two textual descriptions.
- **UFine6926** [48] is a recently constructed high quality dataset with ultra-fine granularity, which contains 26,206 images from 6,926 identities, and each image has two textual descriptions.

C More Qualitative Analysis

C.1 Visual Token Selection Visualization

To better understand how our model selects informative visual tokens, we present more visualization results of token selection on RSTPReid with varying selection ratios $R_s \in \{0, 0.3, 0.4, 0.5, 0.6, 0.7\}$. The figure includes texts, original images, and their corresponding processed outputs, where masked regions highlight tokens identified as redundant and subsequently discarded. As shown in Fig.7, when $R_s = 0$, all visual tokens are retained, resulting in no suppression of background noise. As the selection ratio gradually increases, redundant or semantically irrelevant regions, like the background road and vehicles, are progressively discarded. Notably, at moderate ratios like $R_s = 0.4$ and $R_s = 0.5$, the model successfully suppresses the distracting “car” region while preserving key semantic regions such as the “black jacket”, “blue jeans”, and “bag”, which directly correspond to the textual query. This behavior illustrates the model’s capacity to identify and retain visual tokens that are most semantically aligned with the query. Excessively high selection ratios (e.g., $R_s = 0.6$ or $R_s = 0.7$) begin to preserve redundant content, which may impair alignment. Overall, these visualizations highlight the effectiveness of our token selection mechanism in suppressing background noise and focusing attention on semantically salient regions. By identifying and retaining the most informative visual tokens, the model not only enhances alignment accuracy but also improves interpretability and robustness in complex visual scenes.

C.2 Visual Token Merging Visualization

To further interpret how our model merges informative visual tokens, we provide more visualizations of token merging results under varying merging ratios $R_m \in \{0.05, 0.1, 0.3, 0.5\}$. For clearer visualization, the selected visual tokens are displayed as colored masks in Fig.8, where distinct colors assigned by the DPC-KNN algorithm indicate token groupings. Tokens sharing the same color belong to the same cluster, reflecting their semantic and spatial coherence. These results demonstrate that our merging strategy successfully consolidates fine-grained tokens (e.g., “jacket”, “shoes”, and “jeans”) into semantically meaningful clusters, improving the completeness and expressiveness of the representation. Moreover, DFSM can effectively cluster attribute tokens of different visual scales (e.g., “jacket” and “shoes”) under appropriate merging ratios, reflecting its ability to maintain semantic coherence while adapting diverse granularity levels. However, as R_m increases beyond an appropriate level, the merging becomes overly aggressive, potentially suppressing fine-grained semantic distinctions and reducing the overall representational fidelity.

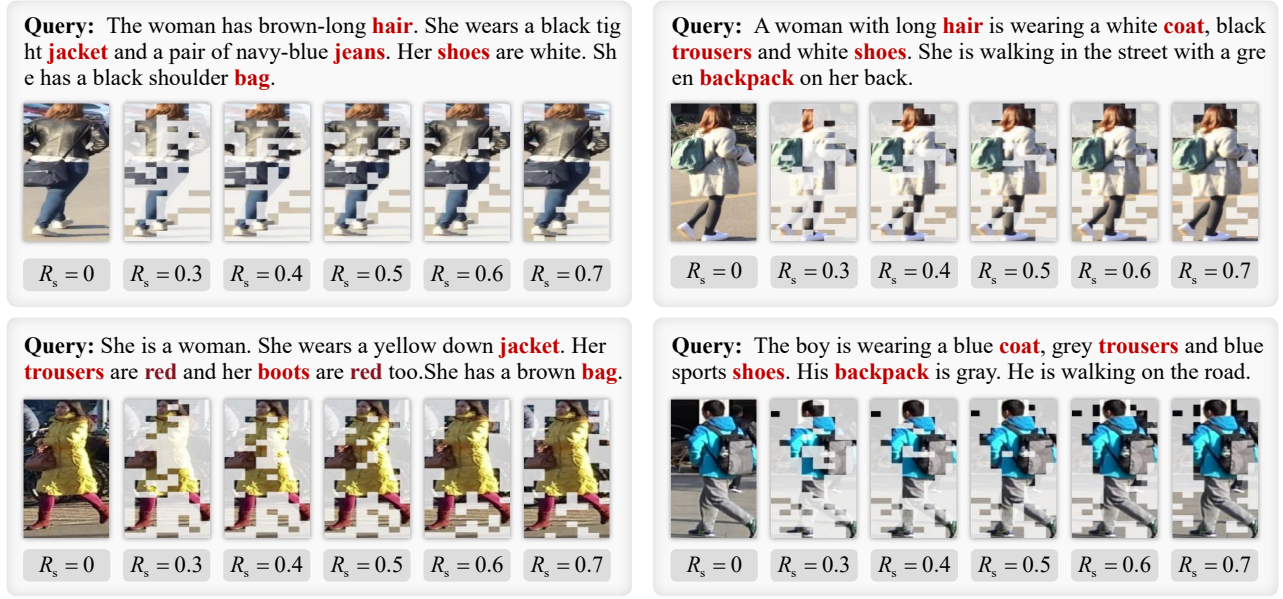


Figure 7: The visualization of selected tokens with associated texts under different selection ratios.

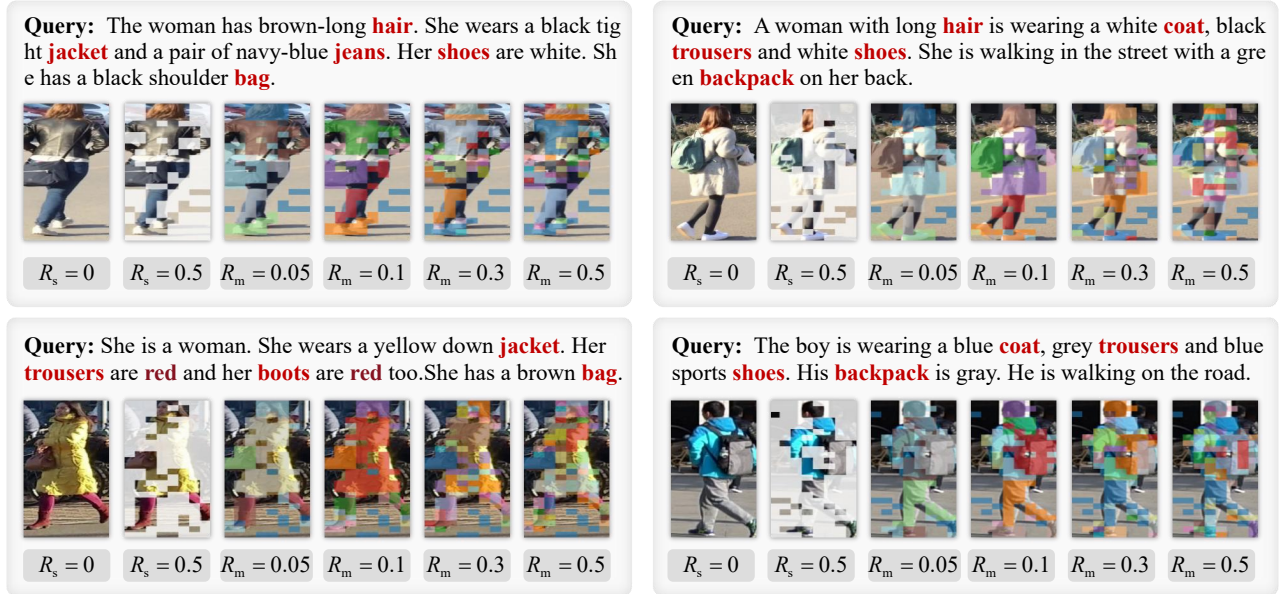


Figure 8: The visualization of merged tokens with the different merging ratios.

C.3 Region-Word Alignment Visualization

To qualitatively assess the fine-grained cross-modal alignment ability of our model, we provide more region-word alignment visualizations in Fig.9. Specifically, we illustrate attention maps that highlight the associations between individual words in the textual query and corresponding image regions, where colors closer to red reflect more semantic relevance. As shown in Fig.9, DFSM accurately attends to semantic regions corresponding to attribute words such as “backpack”, “jacket”, and “coat”. Even for small or

less visually prominent concepts like “shoes”, the model produces highly focused and precise attention, indicating strong fine-grained modeling capabilities. In addition to clothing and body-part attributes, DFSM is also capable of capturing color-related cues such as “red” and “gray”, and attends to all relevant regions rather than isolated instances, demonstrating comprehensive attribute understanding. These visualizations show that DFSM not only captures identity-relevant fine-grained features but also establishes localized semantic correspondences across modalities.

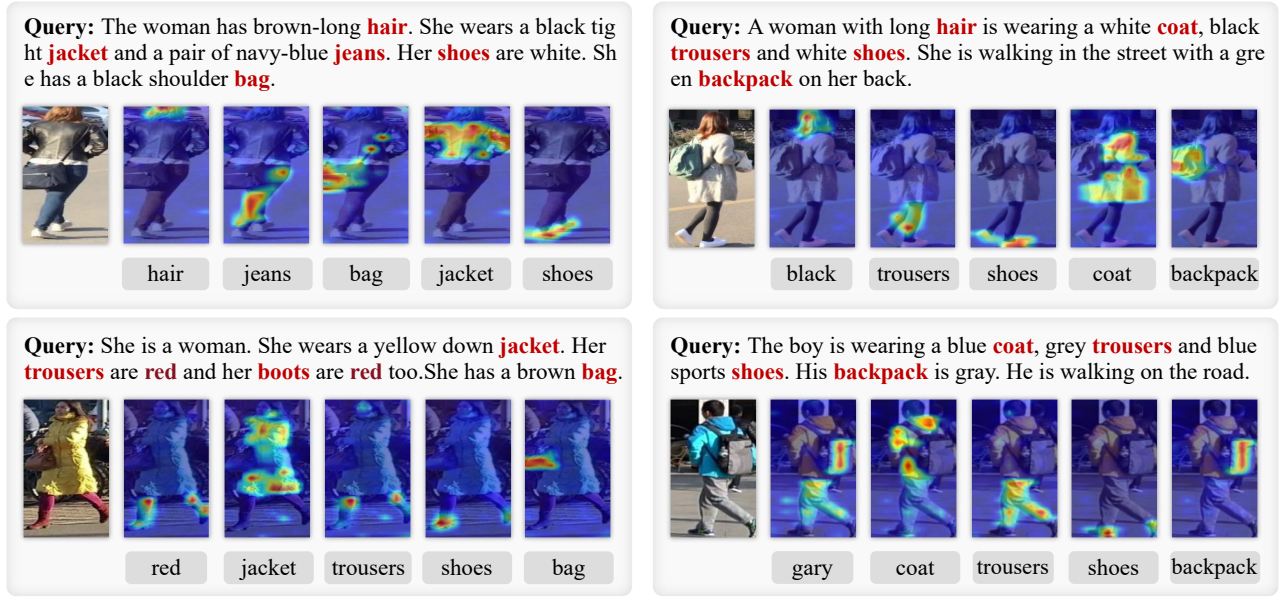


Figure 9: The visualization of region-word alignment. Highlighted visual regions indicate relevance to the words.

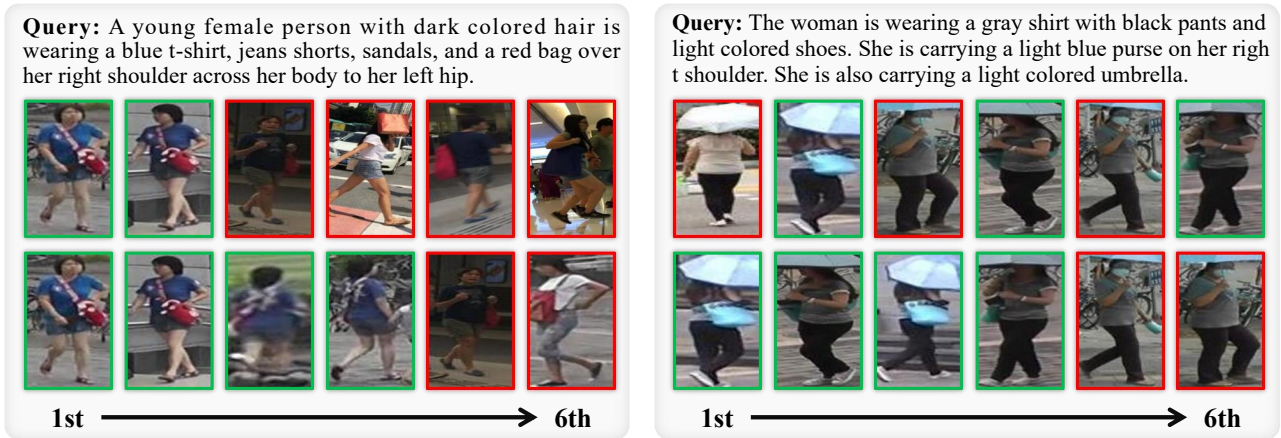


Figure 10: Comparison of top-6 retrieval results between baseline (the first row) and DFSM (the second row). Green and red boxes indicate correct and incorrect matching results, respectively.

C.4 Retrieval Results Visualization

To further assess the practical effectiveness of our proposed DFSM in TPR, we present top-6 retrieval examples in Fig.10, comparing results between the baseline model (first row) and DFSM (second row) under the same query. DFSM consistently retrieves semantically relevant instances, even in diverse and challenging scenarios involving visually similar identities. As shown in Fig.10 (left), while the baseline model captures some key attributes such as the “red bag” and “blue t-shirt” in the top-ranked results, subsequent retrieved images deviate from the core semantics, with several failing to reflect these distinctive features. This indicates limited semantic consistency and

ranking robustness. In contrast, DFSM consistently retrieves images aligned with the text, demonstrating stronger visual-semantic correspondence and enhanced retrieval stability.

In addition, compared to the baseline, DFSM demonstrates superior precision in capturing subtle semantic cues. For instance, in the example on the right of Fig.10, DFSM accurately distinguishes between pedestrians based on fine-grained attributes such as the presence of a blue bag, which are often overlooked by the baseline. This enhanced discriminative capability enables DFSM to retrieve more accurate and diverse results. Notably, the top four retrieved images successfully correspond to the correct identity, highlighting the model’s robustness in resolving semantic ambiguities and its enhanced discriminative power under challenging scenarios.